



TECHNICAL DATA SHEET

Square Hollow Section (SHS) — 200 mm × 200 mm × 3 mm

Description: Cold-formed or hot-finished structural steel square tube with a nominal profile of 200 mm by 200 mm and a uniform 3 mm wall thickness. Widely utilized in structural frameworks, machine construction, civil engineering, and general architecture due to its symmetrical strength distribution.

1. Dimensional & Geometric Properties

Property	Metric Value	Imperial Unit Equivalent
Outer Width × Height	200 mm × 200 mm	7.874 in × 7.874 in
Wall Thickness (t)	3.0 mm	0.118 in (approx. 11 Gauge)
Nominal Mass per Meter	~18.3 kg/m	~12.3 lbs/ft
Cross-Sectional Area (A)	23.3 cm ²	3.61 in ²
Moment of Inertia (I _x / I _y)	1480 cm ⁴	35.56 in ⁴
Section Modulus (W _{el})	148 cm ³	9.03 in ³
Radius of Gyration (i)	7.96 cm	3.13 in

2. Manufacturing Standards & Material Grades

This profile is generally manufactured in compliance with one or more of the following international regulatory frameworks:

Standard	Common Available Grades	Application Environment
EN 10210 / EN 10219	S235JRH, S275J0H, S355J2H	European structural applications, high load profiles
ASTM A500	Grade A, Grade B, Grade C	North American structural framing, welded, bolted frames
AS/NZS 1163	C250L0, C350L0	Australian/Oceania structural/mechanical load tasks
JIS G3466	STKR400, STKR490	Japanese engineering and civil structures





3. Mechanical & Chemical Characteristics (Typical)

Values mapped below reflect common high-strength variations (e.g., S355J2H or ASTM A500 Grade C):

Property Description	Typical Limit (Metric)	Typical Limit (Imperial)
Minimum Yield Strength (R _e)	355 MPa	51,500 psi
Ultimate Tensile Strength (R _m)	470 – 630 MPa	68,000 – 91,300 psi
Minimum Elongation (A ₅)	20%	20%
Carbon Equivalent Value (CEV max)	0.45%	0.45%

4. Standard Tolerances & Surface Customization Options

- **Outside Dimensions:** $\pm 1\%$ with a minimum deviation threshold of ± 0.5 mm.
- **Wall Thickness (t):** $\pm 10\%$ variation limit depending on cold-formed/hot-finished specifics.
- **Squareness of Side:** $90^\circ \pm 1^\circ$ configuration limit.
- **Straightness Deviation:** Max 0.15% of the total cumulative length of the profile.
- **Surface Supply Options:** Black bare finish, light protective anti-rust oiling, Hot-Dip Galvanized (HDG), or mill varnished priming.

